



Automated Segmentation of Cellular Autofluorescence Images with a Convolutional Neural Network

Alexander Soto², Malavika Nidhi¹, Alan Woessner¹, Mamello Mohale¹, Olivia Kolenc¹, Kyle P. Quinn¹
¹Department of Biomedical Engineering, University of Arkansas, Fayetteville, AR 72764, ²Stanford University



Introduction

- NADH and FAD autofluorescence imaging is an emerging non-destructive microscopy technique capable of providing quantitative metabolic information with high spatial and temporal resolution without the need for exogenous stains or dyes. (1)
- Cell-specific information on redox state and cofactor protein binding can be extracted from maps of the optical redox ratio of FAD/(NADH+FAD) autofluorescence intensity and NADH fluorescence lifetime imaging (FLIM), respectively. (1, 2)
- These quantitative measures have been used to understand cell metabolism in various fields, including cancer research and regenerative medicine.
- However, quantitative analysis of cell redox state or FLIM requires accurate segmentation of cellular regions of interest (ROI), and masks defining the ROI are typically generated by manually tracing the region of interest, which is time-consuming and subjective. (3, 4, 6)
- Convolutional neural networks (CNNs) could automate cell segmentation because of their extraordinary ability to detect features and classify images. (5, 6)

Overall Objective: Train a convolutional neural network to segment a range of cell types from different acquisition settings (traditional fluorescence intensity images and FLIM images).

Methods

- NADH and FAD autofluorescence images (1024 x 1024 pixels) were summed from 181 fields of view using a mixture of existing data from mesenchymal stem cells, 3T3 fibroblasts, and keratinocytes (Table).
- NADH FLIM images (1024 x 1024 pixels) collected through time-correlated single photon counting were added to the data set (Table).
- Image intensities were resized to 512 x 512 pixels, normalized to range between 0 and 1, and manually traced to create ground truth logical masks.
- A convolutional neural network was designed and trained in MATLAB based on a U-net architecture (Fig. 1) with an encoder depth of 4, consisting of convolution blocks of 64, 128, 256, and 512 feature detectors (Figure).
- Training occurred over 50 epochs and with data randomly partitioned into training (70%), validation (20%), and testing (10%) sets

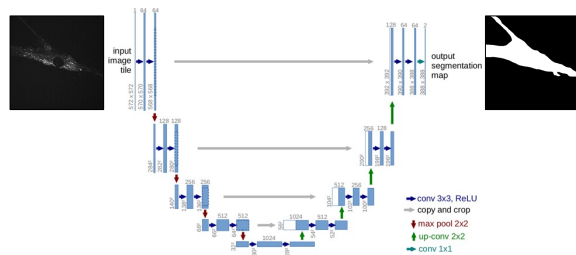


Fig. 1 U-Net CNN architecture used in training. Adapted from (3).

Results

- The trained CNN segmented images in the training set with 89% accuracy, validation with 88%, and testing with 86% with comparable segmentation accuracy across both typical autofluorescence intensity images and FLIM data (Table 1).
- On average, specificity was higher than sensitivity, which meant that the network correctly identified true background pixels at a higher rate than true cell pixels (Table 1).
- Intensity images were overlaid with their ground truth images and network segmented masks to more subjectively critique the network's performance (Fig. 2, 3).

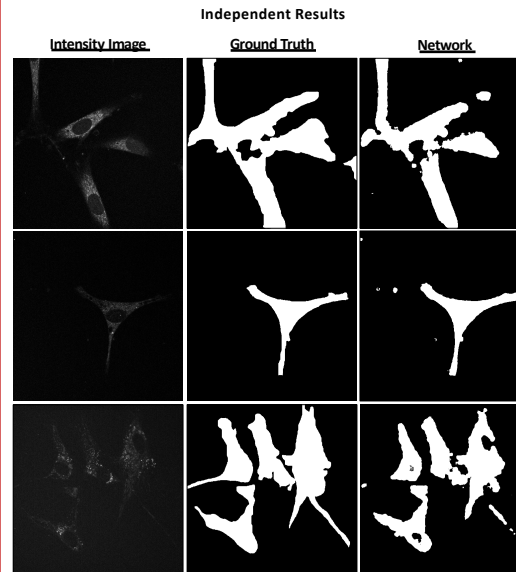


Fig. 2 Results from the validation and testing sets, which were independent of the network's training.

References

- (1) Kolenc, O., Quinn, K. P. (2019) *Evaluating cell metabolism through autofluorescence imaging of NAD(P)H and FAD* (2) Georgakoudi, I., Quinn, K. P. (2023) *Label-Free Optical Metabolic Imaging in Cells and Tissues* (3) Jones, J. D., Ramser, H. E., Woessner, A. E., & Quinn, K. P. (2018) *Communications biology* (4) Woessner, A. E., & Quinn, K. P. (2022) *J biophotonics* (5) Ronneberger, O., Fischer, P., & Brox, T. (2015) <https://doi.org/10.48550/arXiv.1505.04597> (6) Jones, J. D., Rodriguez, M. R., Quinn, K. P. (2021) *Automated Extraction of Skin Wound Healing Biomarkers From In Vivo Label-Free Multiphoton Microscopy Using Convolutional Neural Networks*

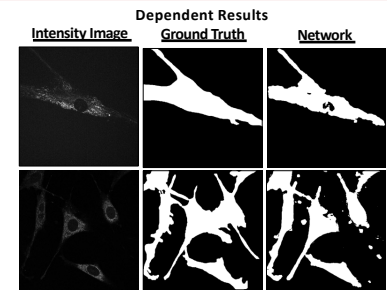


Fig. 3 Results from the training set.

	Accuracy	Sensitivity	Specificity
Training	89%	82%	93%
Validation	88%	80%	93%
Testing	86%	84%	88%

Table 1. Accuracy, sensitivity, and specificity values for each individual dataset.

Discussion

- The network was able to segment intensity images with adequate accuracy in a matter of minutes as opposed to the hours needed for manual segmentation.
- The discrepancy between specificity and sensitivity could be the result of the dataset's uneven class distribution that favored background pixels.
- Moving forward, the network will be retrained on small patches of the original dataset to obtain a balanced class distribution and improve the network's ability to differentiate between cell and background.

Acknowledgements

I would like to sincerely thank Dr. Kyle Quinn, his entire lab, and the Biomedical Engineering REU program for giving me the opportunity to use machine learning for the advancement of biomedical research. Through this exciting experience, I discovered new passions, which will undoubtedly shape my future endeavors. This research was funded by NIH R01EB031032, NIH P20GM139768, and NSF 1846853.